$$
\begin{aligned}
\mathrm{Tr}_{i,j}^{k+o}(f) = \langle \mathsf{st}, \mathbf{A}_f + \mathbf{B}_f^{[:,i+1:i+k+o]}\mathbf{E}\mathbf{C}_f^{[j+1:j+k+o,:]}, \\
\mathbf{B}_f^{[:,1:i]} + \mathbf{B}_f^{[:,i+1:i+k+o]}\mathbf{E}\mathbf{D}_f^{[j+1:j+k+o,1:i]}, \\
\mathbf{C}_f^{[1:j,:]} + \mathbf{D}_f^{[1:j,i+1:i+k+o]}\mathbf{E}\mathbf{C}_f^{[j+1:j+k+o,:]}, \\
\mathbf{D}_f^{[1:j,1:i]} + \mathbf{D}_f^{[1:j,i+1:i+k+o]}\mathbf{E}\mathbf{D}_f^{[j+1:j+k+o,1:i]}\rangle,
\end{aligned}
$$